5. Text Generation with Language Models

CSC3043S Reasoning in AI: Large Language Models

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Some slides from Antoine Bosselut and Yejin Choi.

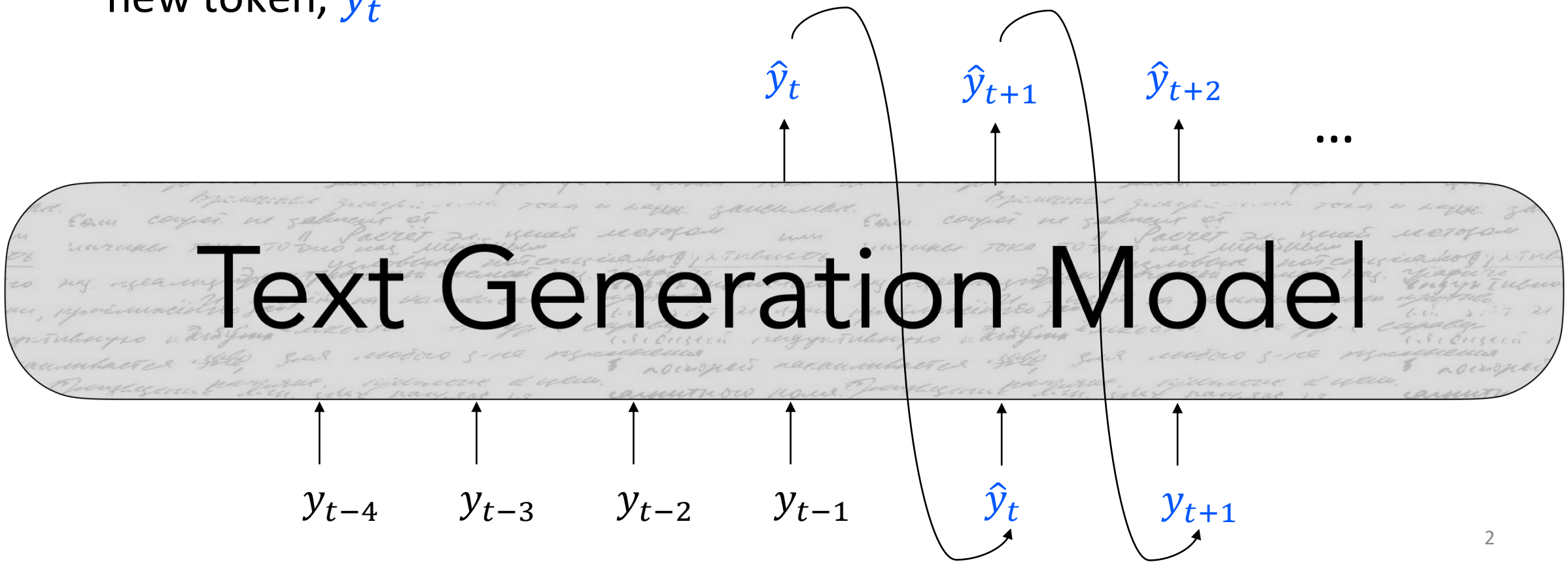


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Text generation

- In autoregressive language models, at each time step *t*, our model takes in a sequence of tokens of text as input $\{y\}_{<t}$ and outputs a new token, $\hat{y}_t$



Text generation

- At each time step *t,* our model computes a vector of scores (called logits) for each token in our vocabulary, $S \in \mathbb{R}^V$:

$$S = f(\{y_{<t}\}, \theta)$$

$f(.)$ is your model

- Then, we compute a probability distribution P over $w \in V$ using these scores:

$$P(y_t = w|\{y_{<t}\}) = \frac{\exp(S_w)}{\sum_{w' \in V} \exp(S_{w'})}$$

Text generation

- We train the model to minimize the negative log likelihood of predicting the next token in the sequence:

$$\mathcal{L}_t = -\log P(y_t^* | \{y_{<t}^*\})$$

Sum $\mathcal{L}_t$ for the entire sequence

- Training with "teacher forcing": feed the observed (not predicted) next token as input at each time step
- At inference time, our **decoding algorithm** defines a function to select a token from this distribution:

$$\hat{y}_t = g(P(y_t | \{y_{<t}\}))$$

$g(.)$ is your decoding algorithm

Decoding

- There are different decoding algorithms for text generation from language models
- The choice of algorithm depends in part on how much uncertainty there is in the output:
  - In tasks such as machine translation, the content is in the input, so there is relatively little uncertainty about the output context (directed generation)
  - In tasks such as creative text generation, the model has to come up with the content in the output, so there is much more uncertainty (open-ended generation)

Decoding

- Open-ended generation: generate word sequence **x** given context **Y**

$$P(\mathbf{x} \mid \mathcal{Y}) = \prod_i P(x_i \mid \mathbf{x}_1^{i-1}, \mathcal{Y})$$



WebText

An unprecedented number of mostly young whales have become stranded on the West Australian coast since 2008.

Poor nutrition has led to a rise in the number of stranded humpback whales on the West Australian coast, veterinary researchers have said. Carly Holyoake, from Murdoch University, at the Australian Veterinary Association's annual conference in Perth on Wednesday, said an unprecedented number of mostly young whales had become stranded on the coast since 2008.

Maximization-based Decoding

- Greedy decoding: Choose the highest scoring word at each time step

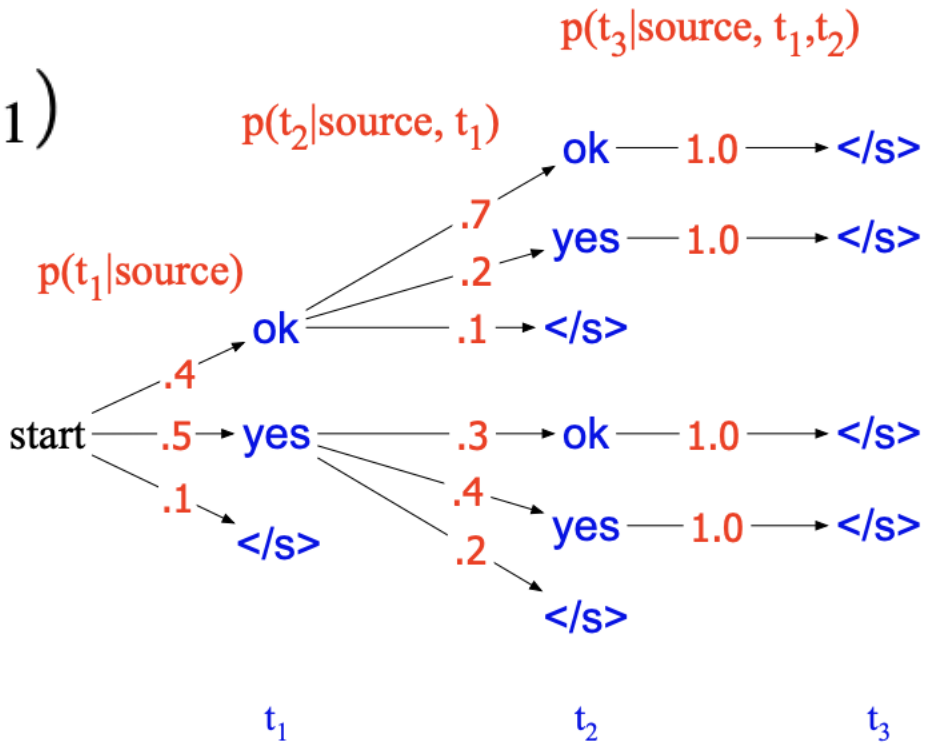
$$x_i = \text{argmax } P(x_i | x_1 \ldots x_{i-1})$$

Maximization-based Decoding

- **Greedy decoding** is a simple approach for generating from the decoder: At each time step, pick the word with the highest probability

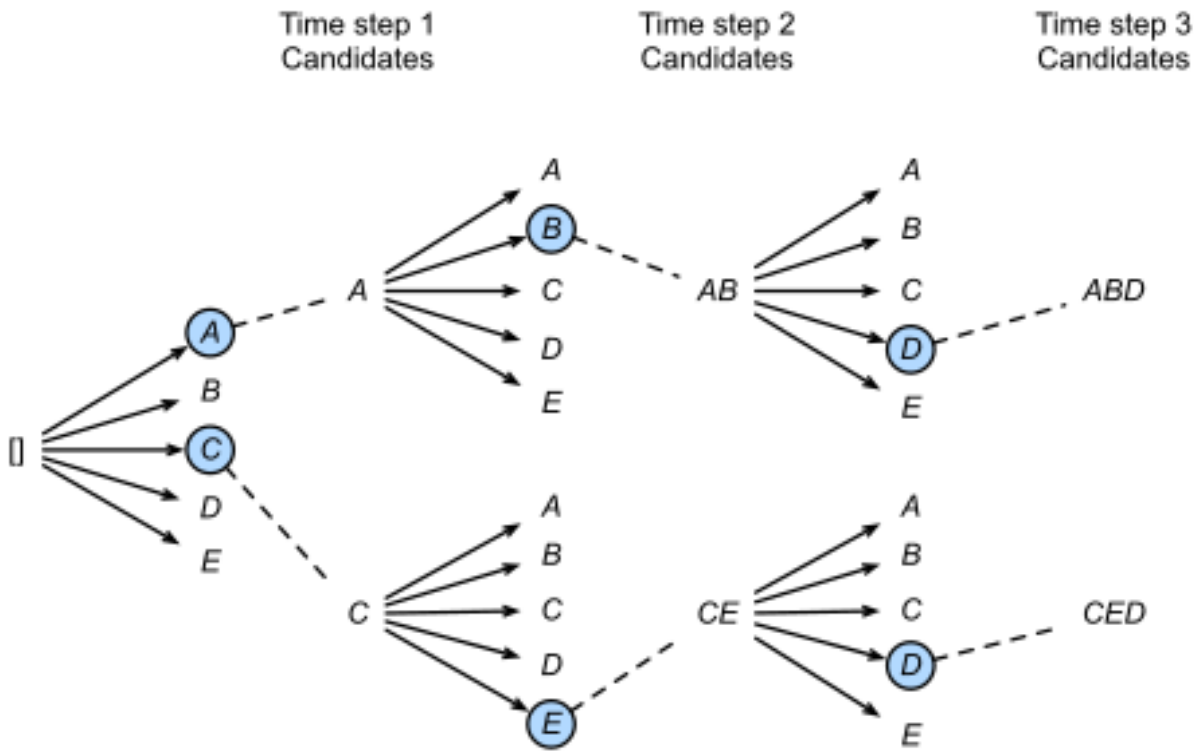
$$\hat{y}_t = \text{argmax}_{w \in V} P(w|x, y_1 \ldots y_{t-1})$$

  - This is not guaranteed to find the globally highest scoring output



Maximization-based Decoding

- Beam search



Beam search

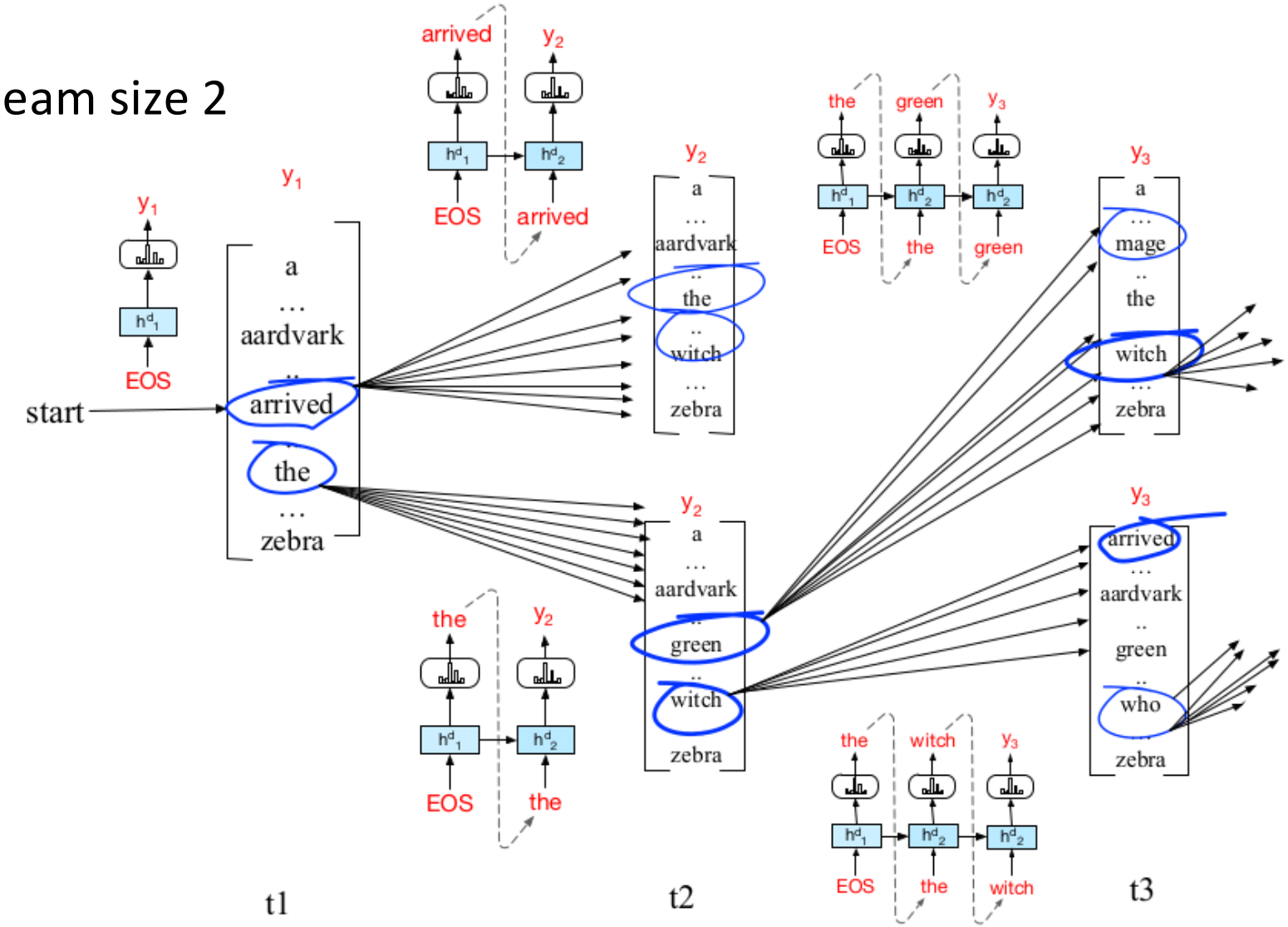
Beam search is an *approximate* search algorithm

- First step: select the *k* highest-scoring options from the output softmax at the first step of the decoder (initial hypotheses)
- For each subsequent step *t*:
  - Compute the softmax for the next output of each of the current hypotheses
  - Prune the $k \cdot V$ expanded hypotheses down to the *k* best hypotheses based on the cumulative score of each hypothesis

$$score(y) = \sum_{i=1}^{t} \log P(y_i | y_1, ..., y_{i-1}, x)$$

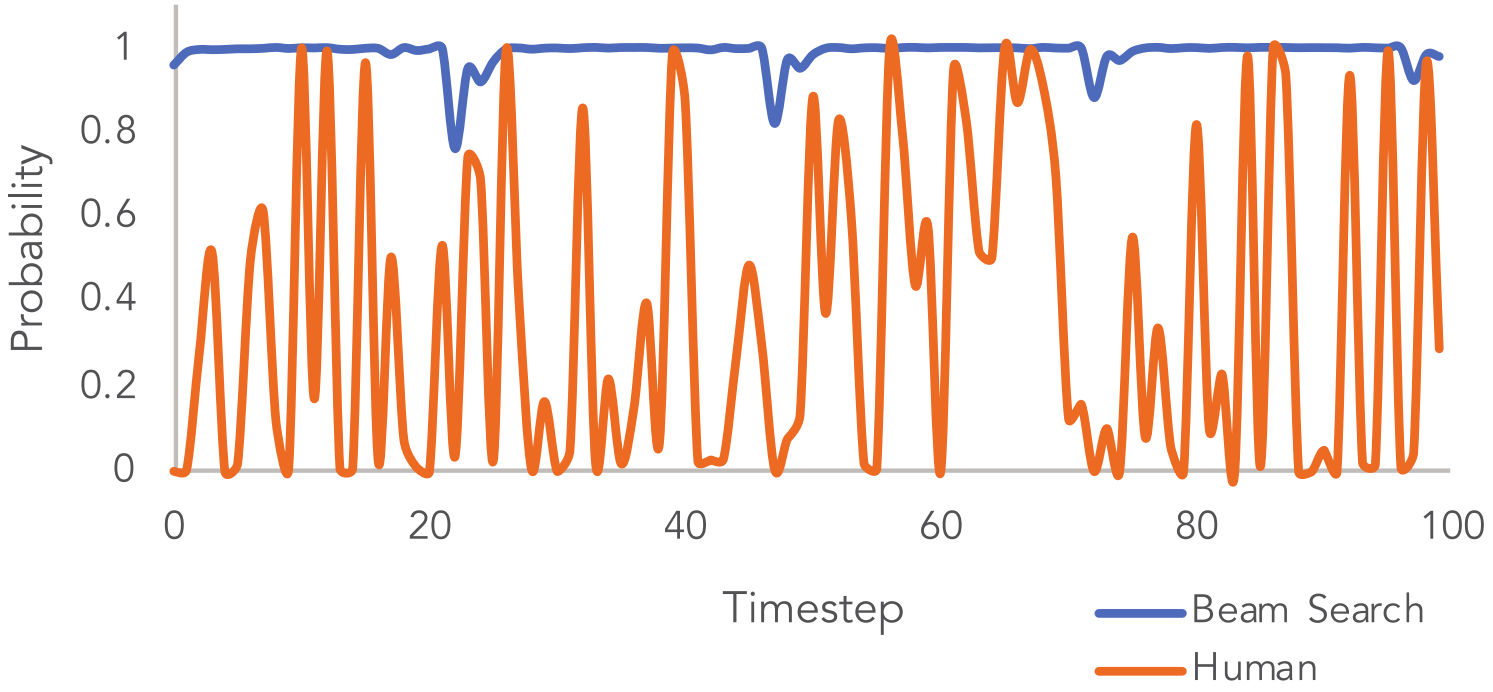
Beam search

- Example: Beam size 2

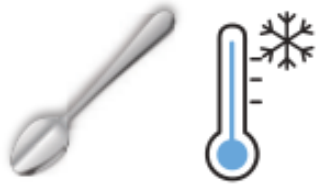


Maximization-based Decoding

- In practice, maximization-based decoding does not generate high-quality output text, e.g. it tends to generate repetitive phrases
- Maximization-based decoding may not reflect how language works



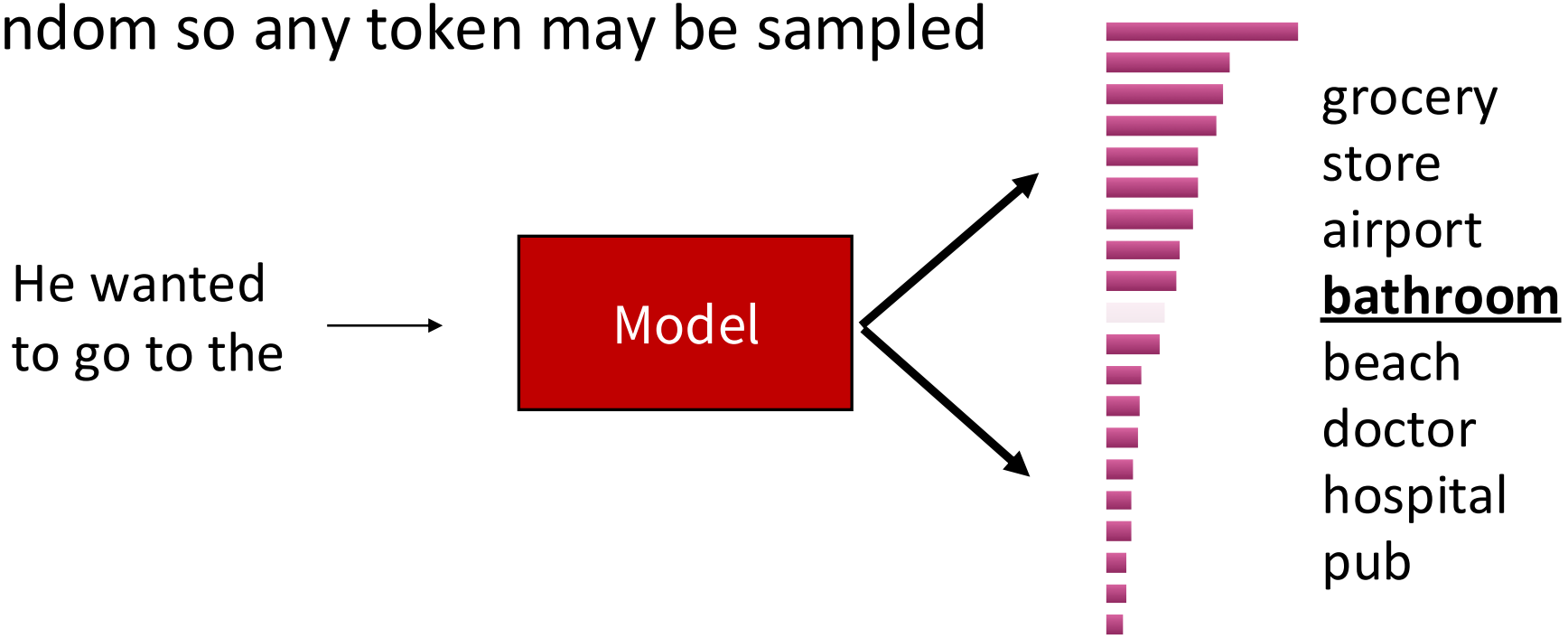
Sampling-based Decoding



- At each time step, sample the next output from the next word distribution

$$\hat{y}_t \sim P(y_t = w \,|\{y\}_{<t})$$

- It is random so any token may be sampled



Sampling-based Decoding

- Problem: Sampling makes every token in the vocabulary an option
  - Even if most of the probability mass in the distribution is over a limited set of options, the tail of the distribution could be very long
  - Many tokens are probably irrelevant in the current context
  - *As a group* there is a high chance than one of these low-probability words will be selected somewhere in the output

- Solution: Truncated sampling: sample only from the *top* words in the probability distribution

Truncated sampling: Top-k Sampling



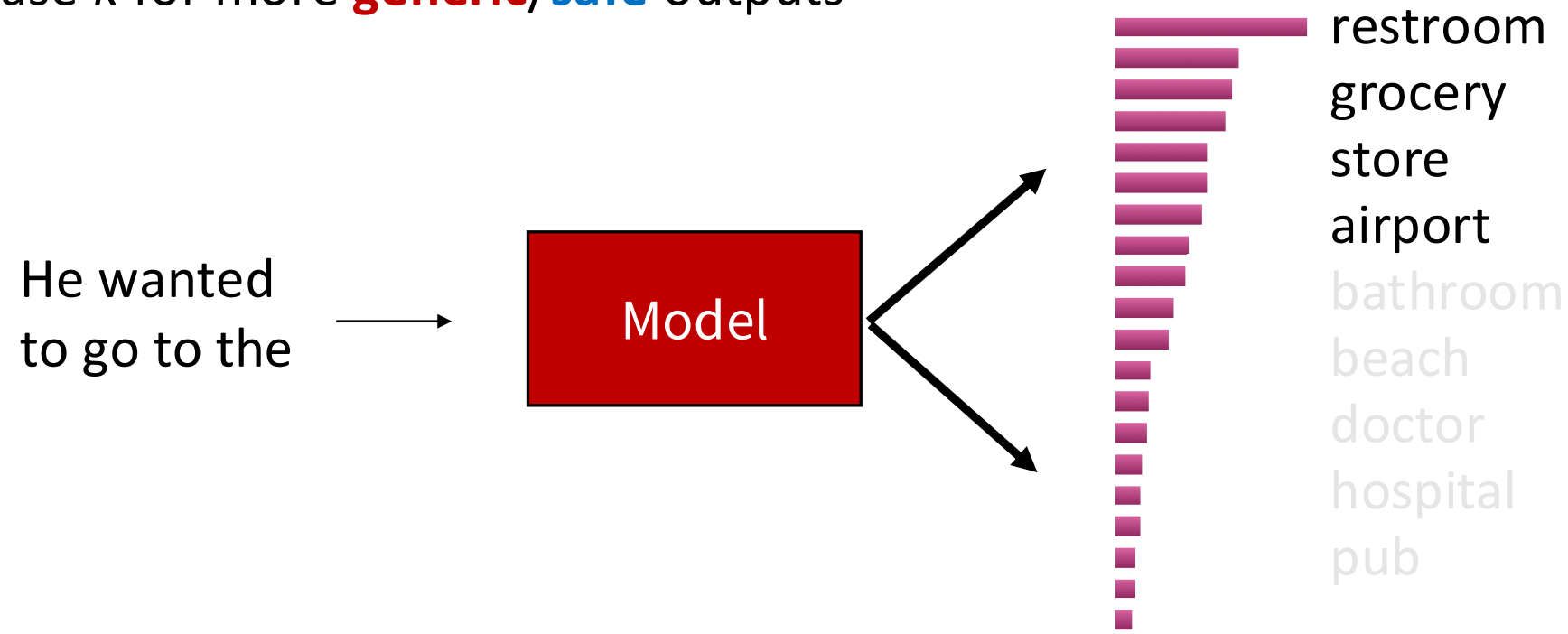
Sample from the top-k next words

Ignore everything else

Fan et al. (2018)

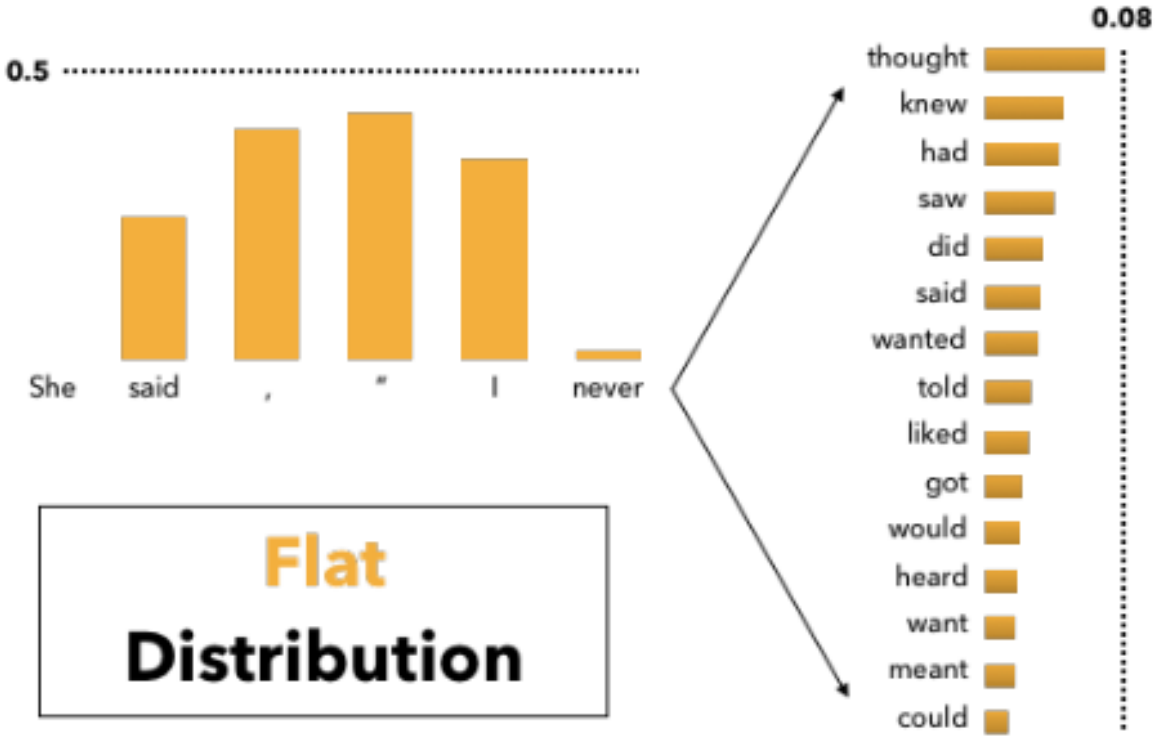
Truncated sampling: Top-k Sampling

- Need to pick the value of *k*
  - Increase *k* for more **diverse**/**risky** outputs
  - Decrease *k* for more **generic**/**safe** outputs



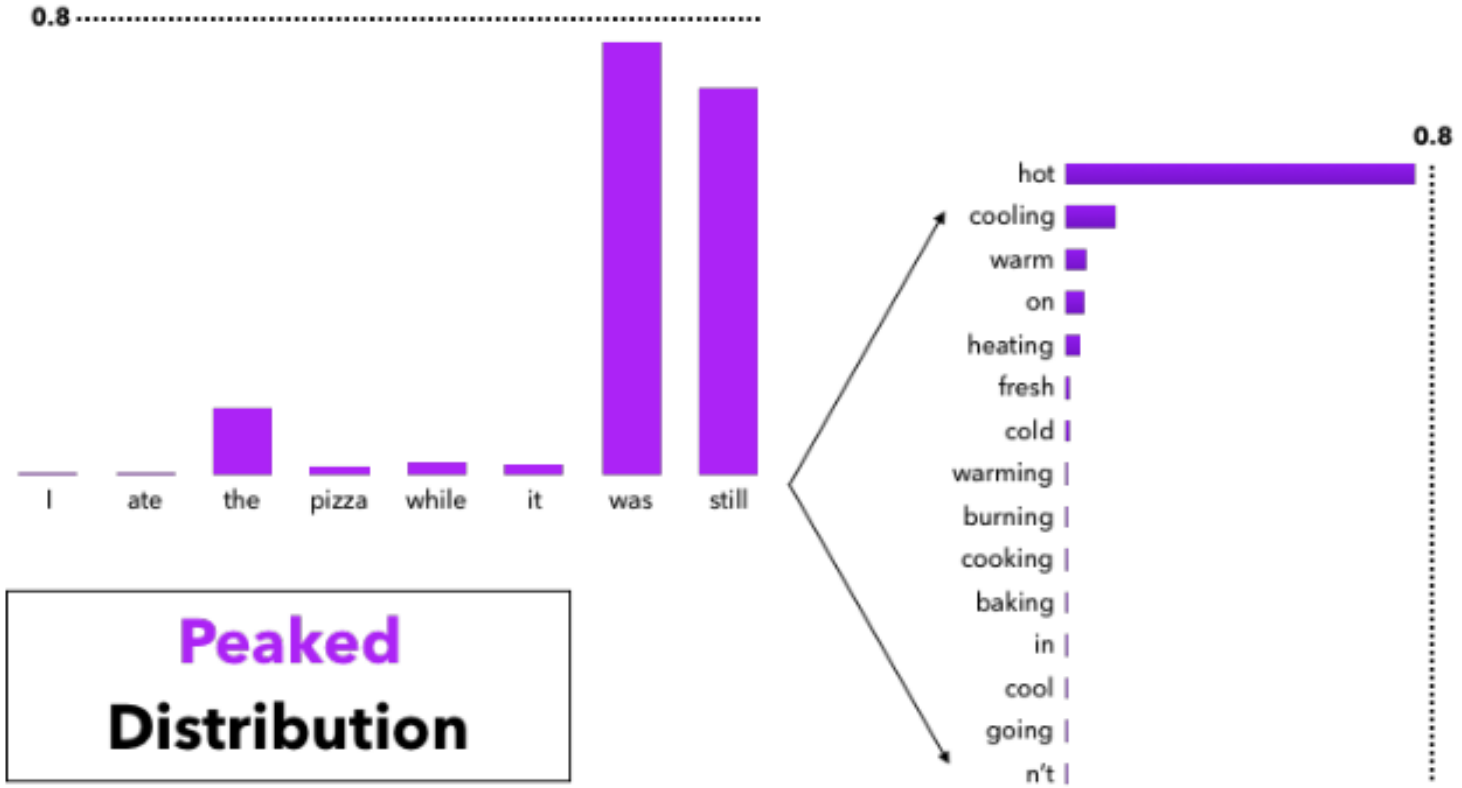
Truncated sampling: Top-k Sampling

- Top-*k* sampling can cut off too quickly



Truncated sampling: Top-k Sampling

- Top-*k* sampling can also cut off too slowly



Truncated sampling: Top-*p* Sampling



Sample from the words with the **top-*p*** probability mass

Ignore everything else

Truncated sampling: Top-*p* Sampling

- Define the top-*p* vocabulary $V^{(p)} \subset V$ as the smallest set such that

$$\underbrace{\sum_{x \in V^{(p)}} P(x|x_{1:i-1})}_{p'} >= p$$

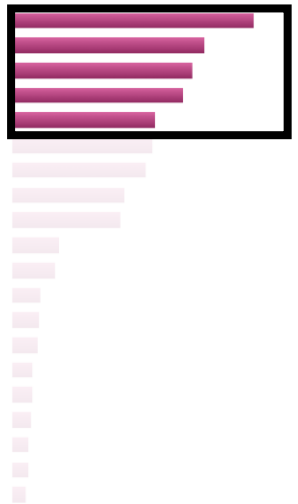
- The distribution sampled from is defined as

$$P'(x|x_{1:i-1}) = \begin{cases} P(x|x_{1:i-1})/p' & \text{if } x \in V^{(k)} \\ 0 & \text{otherwise.} \end{cases}$$

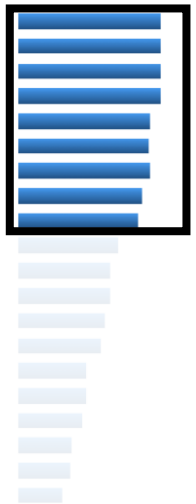
Truncated sampling: Nucleus Sampling

- Top-*p* sampling: The number of words sampled from (*k*) will vary depending on the uniformity of the distribution

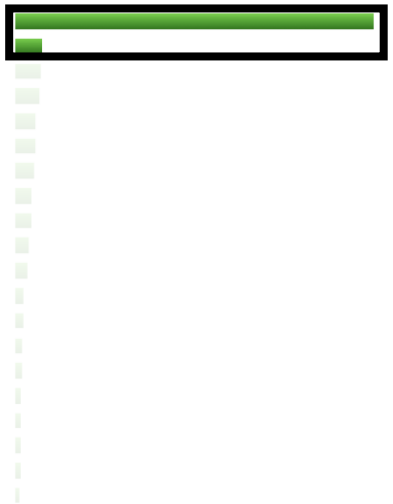
$P_t^1(y_t = w \mid \{y\}_{<t})$



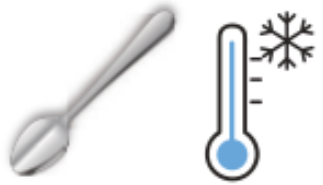
$P_t^2(y_t = w \mid \{y\}_{<t})$



$P_t^3(y_t = w \mid \{y\}_{<t})$



Softmax Temperature



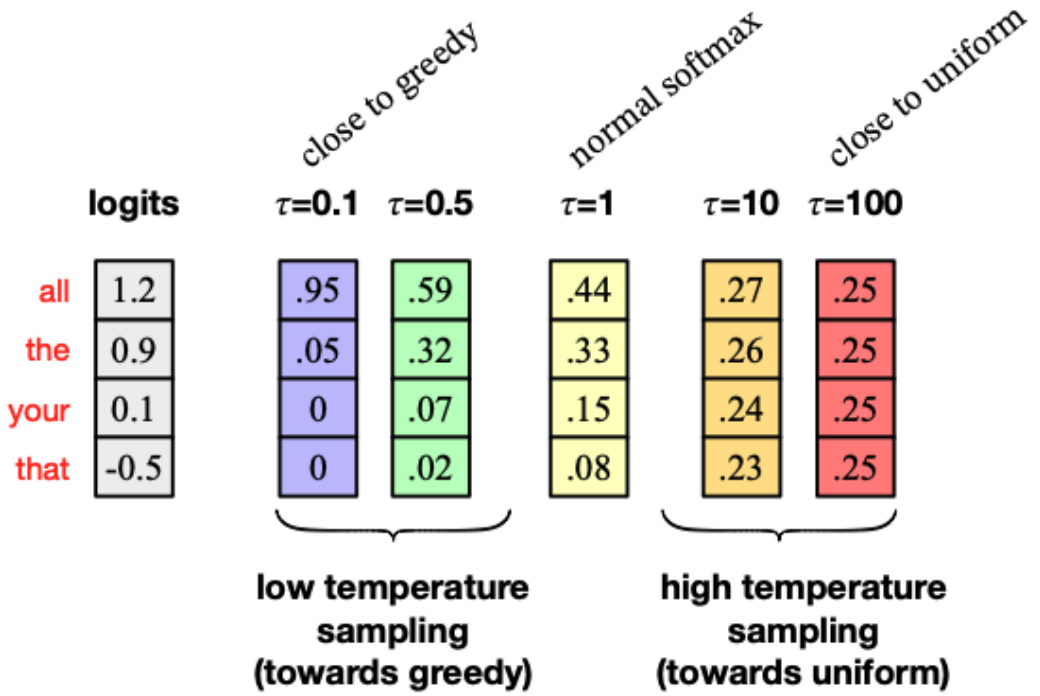
- We can scale the logits before applying the softmax function through a temperature hyperparameter τ – this modifies the shape of the distribution

$$P_t(y_t = w) = \frac{\exp(S_w/\tau)}{\sum_{w' \in V} \exp(S_{w'}/\tau)}$$

- Raise the temperature $\tau > 1$: P_t becomes more uniform
  - More diverse output (probability is spread around vocab)
- Lower the temperature $\tau < 1$: P_t becomes more spiky
  - Less diverse output (probability is concentrated on top words)
- We can apply temperature scaling in conjunction with a decoding algorithm – it is not a decoding algorithm on its own

Softmax Temperature

- Example of effect of temperature τ on softmax output:



Jurafsky and Martin Ch 7 (2026)